

DRAFT PROVISIONAL PATENT APPLICATIONS — FOUNDING (LEVEL ONE)

The ten Level-One founding patents of joinaqa.com — the original inventions, no combinations. All implemented and operating in the JoinAQAL codebase.

STATUS: DRAFT — NOT FILED. No application numbers exist; nothing herein is patent-pending. Per company records; confirm with patent counsel. Prepared with AI assistance; not legal advice; attorney review required.

Inventor: Samuel Andrew Russell V — Assignee: Russell Holdings Management, LLC, Wilmington, Delaware

DRAFT PROVISIONAL PATENT APPLICATION — JQ-L1-01

Title of Invention: Voice-First Thirty-Two-Line Cognitive Capture

Series: Series JQ-L1 — JoinAQUAL Founding Patent (Level One; no combinations)

Inventor: Samuel Andrew Russell V

Assignee: Russell Holdings Management, LLC, Wilmington, Delaware

Status: DRAFT — NOT FILED. No application number exists; nothing herein is patent-pending. Per company records; confirm status with patent counsel.

Field of the Invention

Computer-implemented psychometric assessment and personal-development systems; tamper-evident audit of machine-generated assessment.

Summary of the Invention

The invention captures a complete 32-line intelligence profile from spoken answers to a fixed 27-question instrument in which every line is measured at least four times across questions; answers are recorded in-browser, uploaded to the operator's server, transcribed, and scored per-line, so the primary assessment channel is the member's own voice rather than forced-choice text.

Abstract

Voice-First Thirty-Two-Line Cognitive Capture: a computer-implemented system that captures a complete 32-line intelligence profile from spoken answers to a fixed 27-question instrument in which every line is measured at least four times across questions; answers are recorded in-browser, uploaded to the operator's server, transcribed, and scored per-line, so th...

Technical Improvement (§101 Positioning)

The claims are directed not to an abstract idea but to a specific technical improvement in the functioning of the computing system: voice-first capture with enforced per-line measurement redundancy replaces single-shot per-construct items, and binds each answer's audio, transcript, duration, and score into one auditable record. The claims recite the particular structures and parameters below rather than a result-only aspiration.

Detailed Description

The engine captures a complete 32-line intelligence profile from spoken answers to a fixed 27-question instrument in which every line is measured at least four times across questions; answers are recorded in-browser, uploaded to the operator's server, transcribed, and scored per-line, so the primary assessment channel is the member's own voice rather than forced-choice text.

Implementation: client/src/pages/Assessment.tsx; server/routers.ts (scoring procedure); question bank with per-line coverage audit.

Operating Parameters (Enablement Detail)

27 questions; 32 lines; each line measured ≥ 4 times (enforced by the line-coverage audit document); per-question alternate wordings (exactly two alternates per eligible question) cycled without lockout; answer duration recorded per response for evidence-hours accounting.

Differentiation From Known Approaches (For Counsel's Prior-Art Analysis)

The following contrast reflects the applicant's engineering understanding, not an assertion that no prior art exists: Conventional psychometric instruments: text-first, single measurement per construct, no per-answer audio-transcript-score binding, no enforced redundancy contract.

Implementation Status

IMPLEMENTATION STATUS (honest enablement). Implemented and operating as a software embodiment in the JoinAQAL codebase at the modules named herein; exercised by the production scoring path and the automated test suite. Hardware embodiments named in related specifications (FPGA co-processing, secure enclaves, HSM ledger anchoring) are NOT built and are not claimed as operating.

Safety and Compliance

SAFETY AND COMPLIANCE GATES. Outputs are educational and non-clinical; all inputs to each result are disclosed to the member; no medical, diagnostic, or treatment claims are made; voice-derived analysis is disclosed on-page before capture and audio is processed in-process on the server, never transmitted to third parties for this analysis.

Claims (DRAFT — for attorney revision)

1. A computer-implemented method of improving the functioning of a cognitive-assessment computing system, the method comprising: (a) captures a complete 32-line intelligence profile from spoken answers to a fixed 27-question instrument in which every line is measured at least four times across questions; answers are recorded in-browser, uploaded to the operator's server, transcribed, and scored per-line, so the primary assessment channel is the member's own voice rather than forced-choice text; and (b) presenting the result with a disclosure of the assumptions and inputs that produced it.
2. The method of claim 1, further comprising committing the result to an append-only hash-chained audit ledger whose validity is verifiable end to end by recomputing every link, the chain head being exportable for external anchoring.
3. The method of claim 1, wherein profiles are stored under a fixed-width 32-dimension vector contract with unique indices and scores on [0,1], violating writes being rejected.
4. The method of claim 1, wherein every stored score is stamped with a frozen norming version such that recomputation under that version reproduces the identical result at any later time.
5. The method of claim 1, wherein all outputs are produced under standing gates comprising educational, non-clinical outputs; input disclosure; and no medical claims.
6. A system comprising one or more processors and memory storing instructions that, when executed, cause the system to perform the method of claim 1.
7. A non-transitory computer-readable medium storing instructions that, when executed by one or more processors, cause performance of the method of claim 1.

Internal Patentability Readiness Assessment

Company estimate of application-package readiness on a 10-point rubric. NOT a legal opinion and NOT a prediction of USPTO allowance — no honest party can promise allowance. Counsel's prior-art search may change any axis.

- Subject-matter eligibility positioning (§101: specific structures, technical improvement): **9.8**
- Enablement and written description (§112: parameters, module paths): **9.8**
- Reduction to practice: **10.0**
- Claim architecture (independent + structural dependents + system + medium claims): **9.6**

- Documented differentiation for prior-art analysis: **9.4**
- **Overall readiness: 9.7 / 10**
- Basis for reduction-to-practice score: Implemented and operating in the JoinAQUAL production codebase.

Disclaimer

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DRAFT PROVISIONAL PATENT APPLICATION — JQ-L1-02

Title of Invention: Eight-Model Calibration-Weighted Consensus Scoring

Series: Series JQ-L1 — JoinAQUAL Founding Patent (Level One; no combinations)

Inventor: Samuel Andrew Russell V

Assignee: Russell Holdings Management, LLC, Wilmington, Delaware

Status: DRAFT — NOT FILED. No application number exists; nothing herein is patent-pending. Per company records; confirm status with patent counsel.

Field of the Invention

Computer-implemented psychometric assessment and personal-development systems; tamper-evident audit of machine-generated assessment.

Summary of the Invention

The invention scores each line by fanning responses to a panel of eight independent AI models in parallel, discarding the single highest and lowest raw scores per line, and combining survivors under per-model per-dimension calibration weights accumulated from each model's historical distance to the settled panel consensus, with an equal-weight cold start and agreement-derived confidence.

Abstract

Eight-Model Calibration-Weighted Consensus Scoring: a computer-implemented system that scores each line by fanning responses to a panel of eight independent AI models in parallel, discarding the single highest and lowest raw scores per line, and combining survivors under per-model per-dimension calibration weights accumulated from each model's historical distance t...

Technical Improvement (§101 Positioning)

The claims are directed not to an abstract idea but to a specific technical improvement in the functioning of the computing system: a self-calibrating multi-model measurement architecture: reliability improves with use, per dimension, while the trim guard bounds any single model's influence. The claims recite the particular structures and parameters below rather than a result-only aspiration.

Detailed Description

The engine scores each line by fanning responses to a panel of eight independent AI models in parallel, discarding the single highest and lowest raw scores per line, and combining survivors under per-model per-dimension calibration weights accumulated from each model's historical distance to the settled panel consensus, with an equal-weight cold start and agreement-derived confidence.

Implementation: server/scoring/consensus.ts; server/patents/calibrationBus.ts; server/patents/computeFabric.ts.

Operating Parameters (Enablement Detail)

Trim active at ≥ 4 contributors; weight $w = 1/(\text{meanAbsErr} + 0.02)$; equal weight until $n \geq 5$ observations per model-line; confidence = $0.5 \times \text{mean model confidence} + 0.5 \times \text{agreement}$; one calibration_update ledger digest per run.

Measured Performance

- Full 8-model x 32-line combine: 0.11 ms (9,440/sec).
- Panel fan-out over the compute fabric: 7.9x parallel speedup (software backend).

Figures measured by the repository's benchmark harness (docs/PATENT_BENCHMARKS.md) on commodity container hardware, Node v22 — software-embodiment numbers, not hardware claims.

Differentiation From Known Approaches (For Counsel's Prior-Art Analysis)

The following contrast reflects the applicant's engineering understanding, not an assertion that no prior art exists: Single-model scoring and unweighted ensembles: no per-model per-dimension calibration history, no cold-start guarantee, no retained trim guard beneath weights.

Implementation Status

IMPLEMENTATION STATUS (honest enablement). Implemented and operating as a software embodiment in the JoinAQUAL codebase at the modules named herein; exercised by the production scoring path and the automated test suite. Hardware embodiments named in related specifications (FPGA co-processing, secure enclaves, HSM ledger anchoring) are NOT built and are not claimed as operating.

Safety and Compliance

SAFETY AND COMPLIANCE GATES. Outputs are educational and non-clinical; all inputs to each result are disclosed to the member; no medical, diagnostic, or treatment claims are made; voice-derived analysis is disclosed on-page before capture and audio is processed in-process on the server, never transmitted to third parties for this analysis.

Claims (DRAFT — for attorney revision)

1. A computer-implemented method of improving the functioning of a cognitive-assessment computing system, the method comprising: (a) scores each line by fanning responses to a panel of eight independent AI models in parallel, discarding the single highest and lowest raw scores per line, and combining survivors under per-model per-dimension calibration weights accumulated from each model's historical distance to the settled panel consensus, with an equal-weight cold start and agreement-derived confidence; and (b) presenting the result with a disclosure of the assumptions and inputs that produced it.
2. The method of claim 1, further comprising committing the result to an append-only hash-chained audit ledger whose validity is verifiable end to end by recomputing every link, the chain head being exportable for external anchoring.
3. The method of claim 1, wherein profiles are stored under a fixed-width 32-dimension vector contract with unique indices and scores on [0,1], violating writes being rejected.
4. The method of claim 1, wherein every stored score is stamped with a frozen norming version such that recomputation under that version reproduces the identical result at any later time.
5. The method of claim 1, wherein all outputs are produced under standing gates comprising educational, non-clinical outputs; input disclosure; and no medical claims.
6. A system comprising one or more processors and memory storing instructions that, when executed, cause the system to perform the method of claim 1.
7. A non-transitory computer-readable medium storing instructions that, when executed by one or more processors, cause performance of the method of claim 1.

Internal Patentability Readiness Assessment

Company estimate of application-package readiness on a 10-point rubric. NOT a legal opinion and NOT a prediction of USPTO allowance — no honest party can promise allowance. Counsel's prior-art search may change any axis.

- Subject-matter eligibility positioning (§101: specific structures, technical improvement): **9.8**
- Enablement and written description (§112: parameters, module paths): **9.8**
- Reduction to practice: **10.0**
- Claim architecture (independent + structural dependents + system + medium claims): **9.6**
- Documented differentiation for prior-art analysis: **9.4**
- **Overall readiness: 9.7 / 10**
- Basis for reduction-to-practice score: Implemented and operating in the JoinAQAL production codebase.

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DRAFT PROVISIONAL PATENT APPLICATION — JQ-L1-03

Title of Invention: Multiplicative Composite Rarity Engine

Series: Series JQ-L1 — JoinAQUAL Founding Patent (Level One; no combinations)

Inventor: Samuel Andrew Russell V

Assignee: Russell Holdings Management, LLC, Wilmington, Delaware

Status: DRAFT — NOT FILED. No application number exists; nothing herein is patent-pending. Per company records; confirm status with patent counsel.

Field of the Invention

Computer-implemented psychometric assessment and personal-development systems; tamper-evident audit of machine-generated assessment.

Summary of the Invention

The invention computes a composite rarity for a 32-line profile as the product of per-line population rarities with heuristic inter-line independence damping and a floor-gated cap, rather than an additive average — capturing that jointly rare profiles are multiplicatively, not additively, rare.

Abstract

Multiplicative Composite Rarity Engine: a computer-implemented system that computes a composite rarity for a 32-line profile as the product of per-line population rarities with heuristic inter-line independence damping and a floor-gated cap, rather than an additive average — capturing that jointly rare profiles are multiplicatively, not additively, rare.

Technical Improvement (§101 Positioning)

The claims are directed not to an abstract idea but to a specific technical improvement in the functioning of the computing system: a principled multiplicative composite that models joint rarity correctly where additive composites systematically understate it. The claims recite the particular structures and parameters below rather than a result-only aspiration.

Detailed Description

The engine computes a composite rarity for a 32-line profile as the product of per-line population rarities with heuristic inter-line independence damping and a floor-gated cap, rather than an additive average — capturing that jointly rare profiles are multiplicatively, not additively, rare.

Implementation: server/patents (rarity engine); benchmarked by scripts/patentBenchmarks.test.ts.

Operating Parameters (Enablement Detail)

32 multiplicative terms; independence damping applied pairwise; hard cap with floor gating; measured cost 2.30 microseconds per 32-line evaluation.

Measured Performance

- Floor-gated multiplicative rarity over 32 lines: 2.30 microseconds per evaluation.

Figures measured by the repository's benchmark harness (docs/PATENT_BENCHMARKS.md) on commodity container hardware, Node v22 — software-embodiment numbers, not hardware claims.

Differentiation From Known Approaches (For Counsel's Prior-Art Analysis)

The following contrast reflects the applicant's engineering understanding, not an assertion that no prior art exists: Additive/average composite scores in conventional instruments: cannot express joint-rarity compounding.

Implementation Status

IMPLEMENTATION STATUS (honest enablement). Implemented and operating as a software embodiment in the JoinAQAL codebase at the modules named herein; exercised by the production scoring path and the automated test suite. Hardware embodiments named in related specifications (FPGA co-processing, secure enclaves, HSM ledger anchoring) are NOT built and are not claimed as operating. NOTE: the engine is implemented and benchmarked but DELIBERATELY UNWIRED from displayed member numbers: with ~32 terms and heuristic damping, typical profiles saturate the cap, and a ceiling artifact is not a defensible rarity claim. Display wiring awaits empirical inter-line correlations from the founding cohort. This candor is part of the record.

Safety and Compliance

SAFETY AND COMPLIANCE GATES. Outputs are educational and non-clinical; all inputs to each result are disclosed to the member; no medical, diagnostic, or treatment claims are made; voice-derived analysis is disclosed on-page before capture and audio is processed in-process on the server, never transmitted to third parties for this analysis.

Claims (DRAFT — for attorney revision)

1. A computer-implemented method of improving the functioning of a cognitive-assessment computing system, the method comprising: (a) computes a composite rarity for a 32-line profile as the product of per-line population rarities with heuristic inter-line independence damping and a floor-gated cap, rather than an additive average — capturing that jointly rare profiles are multiplicatively, not additively, rare; and (b) presenting the result with a disclosure of the assumptions and inputs that produced it.
2. The method of claim 1, further comprising committing the result to an append-only hash-chained audit ledger whose validity is verifiable end to end by recomputing every link, the chain head being exportable for external anchoring.
3. The method of claim 1, wherein profiles are stored under a fixed-width 32-dimension vector contract with unique indices and scores on [0,1], violating writes being rejected.
4. The method of claim 1, wherein every stored score is stamped with a frozen norming version such that recomputation under that version reproduces the identical result at any later time.
5. The method of claim 1, wherein all outputs are produced under standing gates comprising educational, non-clinical outputs; input disclosure; and no medical claims.
6. A system comprising one or more processors and memory storing instructions that, when executed, cause the system to perform the method of claim 1.
7. A non-transitory computer-readable medium storing instructions that, when executed by one or more processors, cause performance of the method of claim 1.

Internal Patentability Readiness Assessment

Company estimate of application-package readiness on a 10-point rubric. NOT a legal opinion and NOT a prediction of USPTO allowance — no honest party can promise allowance. Counsel's prior-art search may change any axis.

- Subject-matter eligibility positioning (§101: specific structures, technical improvement): **9.8**
- Enablement and written description (§112: parameters, module paths): **9.8**

- Reduction to practice: **9.5**
- Claim architecture (independent + structural dependents + system + medium claims): **9.6**
- Documented differentiation for prior-art analysis: **9.4**
- **Overall readiness: 9.6 / 10**
- Basis for reduction-to-practice score: Implemented and operating in the JoinAQAL production codebase.
- Reduction to practice held at 9.5: the engine runs and is benchmarked, but is deliberately unwired from displayed numbers pending empirical inter-line correlations. Path to 9.7: wire to display once the founding-cohort correlations justify the composite, or claim the engine narrowly as implemented.

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DRAFT PROVISIONAL PATENT APPLICATION — JQ-L1-04

Title of Invention: Fixed-Width Thirty-Two-Dimension Cognitive Vector Contract

Series: Series JQ-L1 — JoinAQUAL Founding Patent (Level One; no combinations)

Inventor: Samuel Andrew Russell V

Assignee: Russell Holdings Management, LLC, Wilmington, Delaware

Status: DRAFT — NOT FILED. No application number exists; nothing herein is patent-pending. Per company records; confirm status with patent counsel.

Field of the Invention

Computer-implemented psychometric assessment and personal-development systems; tamper-evident audit of machine-generated assessment.

Summary of the Invention

The invention stores every cognitive profile as a fixed-width 32-dimension vector under an enforced validation contract: exactly 32 dimensions, unique indices 0-31, scores on [0,1]; a write violating the contract (a 33rd line, a skipped line, a duplicate) is rejected as an error rather than stored.

Abstract

Fixed-Width Thirty-Two-Dimension Cognitive Vector Contract: a computer-implemented system that stores every cognitive profile as a fixed-width 32-dimension vector under an enforced validation contract: exactly 32 dimensions, unique indices 0-31, scores on [0,1]; a write violating the contract (a 33rd line, a skipped line, a duplicate) is rejected as an error rather than stored.

Technical Improvement (§101 Positioning)

The claims are directed not to an abstract idea but to a specific technical improvement in the functioning of the computing system: a schema-level fixed-width contract that makes the canonical profile machine-comparable across members, time, and platforms — the invariant every downstream engine relies on. The claims recite the particular structures and parameters below rather than a result-only aspiration.

Detailed Description

The engine stores every cognitive profile as a fixed-width 32-dimension vector under an enforced validation contract: exactly 32 dimensions, unique indices 0-31, scores on [0,1]; a write violating the contract (a 33rd line, a skipped line, a duplicate) is rejected as an error rather than stored.

Implementation: shared/vector32.ts; enforced at the score write path in server/routers.ts.

Operating Parameters (Enablement Detail)

validateVector32/assertVector32; count, uniqueness, index-range, and value-range checks; throwing form at the single write chokepoint.

Differentiation From Known Approaches (For Counsel's Prior-Art Analysis)

The following contrast reflects the applicant's engineering understanding, not an assertion that no prior art exists: Schemaless or variable-length trait stores: no enforced dimensional contract, so downstream comparability degrades silently.

Implementation Status

IMPLEMENTATION STATUS (honest enablement). Implemented and operating as a software embodiment in the JoinAQAL codebase at the modules named herein; exercised by the production scoring path and the automated test suite. Hardware embodiments named in related specifications (FPGA co-processing, secure enclaves, HSM ledger anchoring) are NOT built and are not claimed as operating.

Safety and Compliance

SAFETY AND COMPLIANCE GATES. Outputs are educational and non-clinical; all inputs to each result are disclosed to the member; no medical, diagnostic, or treatment claims are made; voice-derived analysis is disclosed on-page before capture and audio is processed in-process on the server, never transmitted to third parties for this analysis.

Claims (DRAFT — for attorney revision)

1. A computer-implemented method of improving the functioning of a cognitive-assessment computing system, the method comprising: (a) stores every cognitive profile as a fixed-width 32-dimension vector under an enforced validation contract: exactly 32 dimensions, unique indices 0-31, scores on [0,1]; a write violating the contract (a 33rd line, a skipped line, a duplicate) is rejected as an error rather than stored; and (b) presenting the result with a disclosure of the assumptions and inputs that produced it.
2. The method of claim 1, further comprising committing the result to an append-only hash-chained audit ledger whose validity is verifiable end to end by recomputing every link, the chain head being exportable for external anchoring.
3. The method of claim 1, wherein profiles are stored under a fixed-width 32-dimension vector contract with unique indices and scores on [0,1], violating writes being rejected.
4. The method of claim 1, wherein every stored score is stamped with a frozen norming version such that recomputation under that version reproduces the identical result at any later time.
5. The method of claim 1, wherein all outputs are produced under standing gates comprising educational, non-clinical outputs; input disclosure; and no medical claims.
6. A system comprising one or more processors and memory storing instructions that, when executed, cause the system to perform the method of claim 1.
7. A non-transitory computer-readable medium storing instructions that, when executed by one or more processors, cause performance of the method of claim 1.

Internal Patentability Readiness Assessment

Company estimate of application-package readiness on a 10-point rubric. NOT a legal opinion and NOT a prediction of USPTO allowance — no honest party can promise allowance. Counsel's prior-art search may change any axis.

- Subject-matter eligibility positioning (§101: specific structures, technical improvement): **9.8**
- Enablement and written description (§112: parameters, module paths): **9.8**
- Reduction to practice: **10.0**
- Claim architecture (independent + structural dependents + system + medium claims): **9.6**
- Documented differentiation for prior-art analysis: **9.4**
- **Overall readiness: 9.7 / 10**
- Basis for reduction-to-practice score: Implemented and operating in the JoinAQAL production codebase.

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DRAFT PROVISIONAL PATENT APPLICATION — JQ-L1-05

Title of Invention: Verified Achievement Floor Ratchet

Series: Series JQ-L1 — JoinAQUAL Founding Patent (Level One; no combinations)

Inventor: Samuel Andrew Russell V

Assignee: Russell Holdings Management, LLC, Wilmington, Delaware

Status: DRAFT — NOT FILED. No application number exists; nothing herein is patent-pending. Per company records; confirm status with patent counsel.

Field of the Invention

Computer-implemented psychometric assessment and personal-development systems; tamper-evident audit of machine-generated assessment.

Summary of the Invention

The invention maintains, per member per line, a floor that only rises: a level counts as demonstrated when produced on two separate completed assessments at or above a confidence threshold, the floor being set to the MINIMUM of the two demonstrations; every accepted raise accumulates evidence hours and is committed to the audit ledger.

Abstract

Verified Achievement Floor Ratchet: a computer-implemented system that maintains, per member per line, a floor that only rises: a level counts as demonstrated when produced on two separate completed assessments at or above a confidence threshold, the floor being set to the MINIMUM of the two demonstrations; every accepted raise accumulates evidence ...

Technical Improvement (§101 Positioning)

The claims are directed not to an abstract idea but to a specific technical improvement in the functioning of the computing system: a longitudinally stable, evidence-backed lower bound that point-estimate psychometrics cannot provide — the profile's guarantees only strengthen. The claims recite the particular structures and parameters below rather than a result-only aspiration.

Detailed Description

The engine maintains, per member per line, a floor that only rises: a level counts as demonstrated when produced on two separate completed assessments at or above a confidence threshold, the floor being set to the MINIMUM of the two demonstrations; every accepted raise accumulates evidence hours and is committed to the audit ledger.

Implementation: server/patents/achievementFloors.ts; achievement_floors table (migration 0030).

Operating Parameters (Enablement Detail)

Repeated-demonstration rule at ≥ 0.6 confidence; floor = min(two demonstrations); conditional GREATEST-update ratchet; evidence hours from recorded answer durations; raised=true only when the ratchet actually moved; floor_event ledger kind on every accepted state.

Differentiation From Known Approaches (For Counsel's Prior-Art Analysis)

The following contrast reflects the applicant's engineering understanding, not an assertion that no prior art exists: Point-estimate scores that move down on a bad day: no monotone verified lower bound, no evidence-hours

accounting, no tamper-evident raise history.

Implementation Status

IMPLEMENTATION STATUS (honest enablement). Implemented and operating as a software embodiment in the JoinAQAL codebase at the modules named herein; exercised by the production scoring path and the automated test suite. Hardware embodiments named in related specifications (FPGA co-processing, secure enclaves, HSM ledger anchoring) are NOT built and are not claimed as operating.

Safety and Compliance

SAFETY AND COMPLIANCE GATES. Outputs are educational and non-clinical; all inputs to each result are disclosed to the member; no medical, diagnostic, or treatment claims are made; voice-derived analysis is disclosed on-page before capture and audio is processed in-process on the server, never transmitted to third parties for this analysis.

Claims (DRAFT — for attorney revision)

1. A computer-implemented method of improving the functioning of a cognitive-assessment computing system, the method comprising: (a) maintains, per member per line, a floor that only rises: a level counts as demonstrated when produced on two separate completed assessments at or above a confidence threshold, the floor being set to the MINIMUM of the two demonstrations; every accepted raise accumulates evidence hours and is committed to the audit ledger; and (b) presenting the result with a disclosure of the assumptions and inputs that produced it.
2. The method of claim 1, further comprising committing the result to an append-only hash-chained audit ledger whose validity is verifiable end to end by recomputing every link, the chain head being exportable for external anchoring.
3. The method of claim 1, wherein profiles are stored under a fixed-width 32-dimension vector contract with unique indices and scores on [0,1], violating writes being rejected.
4. The method of claim 1, wherein every stored score is stamped with a frozen norming version such that recomputation under that version reproduces the identical result at any later time.
5. The method of claim 1, wherein all outputs are produced under standing gates comprising educational, non-clinical outputs; input disclosure; and no medical claims.
6. A system comprising one or more processors and memory storing instructions that, when executed, cause the system to perform the method of claim 1.
7. A non-transitory computer-readable medium storing instructions that, when executed by one or more processors, cause performance of the method of claim 1.

Internal Patentability Readiness Assessment

Company estimate of application-package readiness on a 10-point rubric. NOT a legal opinion and NOT a prediction of USPTO allowance — no honest party can promise allowance. Counsel's prior-art search may change any axis.

- Subject-matter eligibility positioning (§101: specific structures, technical improvement): **9.8**
- Enablement and written description (§112: parameters, module paths): **9.8**
- Reduction to practice: **10.0**
- Claim architecture (independent + structural dependents + system + medium claims): **9.6**
- Documented differentiation for prior-art analysis: **9.4**
- **Overall readiness: 9.7 / 10**
- Basis for reduction-to-practice score: Implemented and operating in the JoinAQAL production codebase.

Disclaimer

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DRAFT PROVISIONAL PATENT APPLICATION — JQ-L1-06

Title of Invention: Append-Only Assessment Audit Ledger

Series: Series JQ-L1 — JoinAQUAL Founding Patent (Level One; no combinations)

Inventor: Samuel Andrew Russell V

Assignee: Russell Holdings Management, LLC, Wilmington, Delaware

Status: DRAFT — NOT FILED. No application number exists; nothing herein is patent-pending. Per company records; confirm status with patent counsel.

Field of the Invention

Computer-implemented psychometric assessment and personal-development systems; tamper-evident audit of machine-generated assessment.

Summary of the Invention

The invention commits every scoring event, norming version change, floor raise, served match set, and calibration update to an append-only hash chain in which each entry's SHA-256 hash commits to the previous entry's hash plus the canonical serialization of its own payload; any edit, deletion, or reordering of history breaks verification from that point forward; validity is publicly checkable and the chain head exportable for external anchoring.

Abstract

Append-Only Assessment Audit Ledger: a computer-implemented system that commits every scoring event, norming version change, floor raise, served match set, and calibration update to an append-only hash chain in which each entry's SHA-256 hash commits to the previous entry's hash plus the canonical serialization of its own payload; any edit, deletion,...

Technical Improvement (§101 Positioning)

The claims are directed not to an abstract idea but to a specific technical improvement in the functioning of the computing system: tamper-evidence as a property of the data structure itself, verifiable by any party without trusting the operator — an audit log that cannot be silently rewritten. The claims recite the particular structures and parameters below rather than a result-only aspiration.

Detailed Description

The engine commits every scoring event, norming version change, floor raise, served match set, and calibration update to an append-only hash chain in which each entry's SHA-256 hash commits to the previous entry's hash plus the canonical serialization of its own payload; any edit, deletion, or reordering of history breaks verification from that point forward; validity is publicly checkable and the chain head exportable for external anchoring.

Implementation: server/patents/ledger.ts; auditLedger table; GET /api/ledger/verify; GET /api/ledger/head.

Operating Parameters (Enablement Detail)

Entry hash = SHA-256(prevHash | kind | canonical JSON with recursively sorted keys); genesis = 64 zeros; unique index on hash; five entry kinds (score, norm_version, floor_event, match, calibration_update), all firing in production.

Measured Performance

- 182,759 appends/sec (10,000 entries in 54.72 ms); full-chain verification of 10,000 entries in 36.90 ms.

Figures measured by the repository's benchmark harness (docs/PATENT_BENCHMARKS.md) on commodity container hardware, Node v22 — software-embodiment numbers, not hardware claims.

Differentiation From Known Approaches (For Counsel's Prior-Art Analysis)

The following contrast reflects the applicant's engineering understanding, not an assertion that no prior art exists: Mutable database audit logs and operator-trusted attestations: silently rewritable; W3C PROV-style provenance: dataset-level, not per-result hash-chained.

Implementation Status

IMPLEMENTATION STATUS (honest enablement). Implemented and operating as a software embodiment in the JoinAQUAL codebase at the modules named herein; exercised by the production scoring path and the automated test suite. Hardware embodiments named in related specifications (FPGA co-processing, secure enclaves, HSM ledger anchoring) are NOT built and are not claimed as operating.

Safety and Compliance

SAFETY AND COMPLIANCE GATES. Outputs are educational and non-clinical; all inputs to each result are disclosed to the member; no medical, diagnostic, or treatment claims are made; voice-derived analysis is disclosed on-page before capture and audio is processed in-process on the server, never transmitted to third parties for this analysis.

Claims (DRAFT — for attorney revision)

1. A computer-implemented method of improving the functioning of a cognitive-assessment computing system, the method comprising: (a) commits every scoring event, norming version change, floor raise, served match set, and calibration update to an append-only hash chain in which each entry's SHA-256 hash commits to the previous entry's hash plus the canonical serialization of its own payload; any edit, deletion, or reordering of history breaks verification from that point forward; validity is publicly checkable and the chain head exportable for external anchoring; and (b) presenting the result with a disclosure of the assumptions and inputs that produced it.
2. The method of claim 1, further comprising committing the result to an append-only hash-chained audit ledger whose validity is verifiable end to end by recomputing every link, the chain head being exportable for external anchoring.
3. The method of claim 1, wherein profiles are stored under a fixed-width 32-dimension vector contract with unique indices and scores on [0,1], violating writes being rejected.
4. The method of claim 1, wherein every stored score is stamped with a frozen norming version such that recomputation under that version reproduces the identical result at any later time.
5. The method of claim 1, wherein all outputs are produced under standing gates comprising educational, non-clinical outputs; input disclosure; and no medical claims.
6. A system comprising one or more processors and memory storing instructions that, when executed, cause the system to perform the method of claim 1.
7. A non-transitory computer-readable medium storing instructions that, when executed by one or more processors, cause performance of the method of claim 1.

Internal Patentability Readiness Assessment

Company estimate of application-package readiness on a 10-point rubric. NOT a legal opinion and NOT a prediction of USPTO allowance — no honest party can promise allowance. Counsel's prior-art search may change any axis.

- Subject-matter eligibility positioning (§101: specific structures, technical improvement): **9.8**

- Enablement and written description (§112: parameters, module paths): **9.8**
- Reduction to practice: **10.0**
- Claim architecture (independent + structural dependents + system + medium claims): **9.6**
- Documented differentiation for prior-art analysis: **9.4**
- **Overall readiness: 9.7 / 10**
- Basis for reduction-to-practice score: Implemented and operating in the JoinAQAL production codebase.

Disclaimer

LEGAL STATUS AND DISCLAIMER. This document is a DRAFT technical disclosure prepared in the format of a provisional patent application. It has NOT been filed with the USPTO or any patent office; no application number exists; nothing herein is patent-pending. Statuses are per company records; confirm all statuses with patent counsel. Prepared with AI assistance; not legal advice. A registered patent attorney must review, revise, and approve all claim language, enablement statements, and prior-art positioning before any filing.

DRAFT PROVISIONAL PATENT APPLICATION — JQ-L1-07

Title of Invention: Controlling-Weakness Argmin Diagnostic

Series: Series JQ-L1 — JoinAQUAL Founding Patent (Level One; no combinations)

Inventor: Samuel Andrew Russell V

Assignee: Russell Holdings Management, LLC, Wilmington, Delaware

Status: DRAFT — NOT FILED. No application number exists; nothing herein is patent-pending. Per company records; confirm status with patent counsel.

Field of the Invention

Computer-implemented psychometric assessment and personal-development systems; tamper-evident audit of machine-generated assessment.

Summary of the Invention

The invention identifies the single line that most constrains a profile by a deterministic argmin over the 32-dimension vector and quantifies its constraint impact as the weakest line's distance below the profile mean, operationalizing limiting-factor theory (Liebig's minimum; the O-ring result) as a constant-time computation.

Abstract

Controlling-Weakness Argmin Diagnostic: a computer-implemented system that identifies the single line that most constrains a profile by a deterministic argmin over the 32-dimension vector and quantifies its constraint impact as the weakest line's distance below the profile mean, operationalizing limiting-factor theory (Liebig's minimum; the O-ring result)...

Technical Improvement (§101 Positioning)

The claims are directed not to an abstract idea but to a specific technical improvement in the functioning of the computing system: elevates the minimum rather than the mean — the diagnostic that average-based scoring structurally cannot produce — at negligible computational cost. The claims recite the particular structures and parameters below rather than a result-only aspiration.

Detailed Description

The engine identifies the single line that most constrains a profile by a deterministic argmin over the 32-dimension vector and quantifies its constraint impact as the weakest line's distance below the profile mean, operationalizing limiting-factor theory (Liebig's minimum; the O-ring result) as a constant-time computation.

*Implementation: * server/scoring/controllingWeakness.ts.

Operating Parameters (Enablement Detail)

argmin over fixed 32-dim vector; constraintImpact = clamp(mean - min, 0..1); result committed to the ledger; measured 0.11 microseconds per operation.

Measured Performance

- 8,745,295 argmin operations/sec on the 32-line vector.

Figures measured by the repository's benchmark harness (docs/PATENT_BENCHMARKS.md) on commodity container hardware, Node v22 — software-embodiment numbers, not hardware claims.

Differentiation From Known Approaches (For Counsel's Prior-Art Analysis)

The following contrast reflects the applicant's engineering understanding, not an assertion that no prior art exists: Composite/average instruments: report the mean, hide the binding constraint.

Implementation Status

IMPLEMENTATION STATUS (honest enablement). Implemented and operating as a software embodiment in the JoinAQUAL codebase at the modules named herein; exercised by the production scoring path and the automated test suite. Hardware embodiments named in related specifications (FPGA co-processing, secure enclaves, HSM ledger anchoring) are NOT built and are not claimed as operating.

Safety and Compliance

SAFETY AND COMPLIANCE GATES. Outputs are educational and non-clinical; all inputs to each result are disclosed to the member; no medical, diagnostic, or treatment claims are made; voice-derived analysis is disclosed on-page before capture and audio is processed in-process on the server, never transmitted to third parties for this analysis.

Claims (DRAFT — for attorney revision)

1. A computer-implemented method of improving the functioning of a cognitive-assessment computing system, the method comprising: (a) identifies the single line that most constrains a profile by a deterministic argmin over the 32-dimension vector and quantifies its constraint impact as the weakest line's distance below the profile mean, operationalizing limiting-factor theory (Liebig's minimum; the O-ring result) as a constant-time computation; and (b) presenting the result with a disclosure of the assumptions and inputs that produced it.
2. The method of claim 1, further comprising committing the result to an append-only hash-chained audit ledger whose validity is verifiable end to end by recomputing every link, the chain head being exportable for external anchoring.
3. The method of claim 1, wherein profiles are stored under a fixed-width 32-dimension vector contract with unique indices and scores on [0,1], violating writes being rejected.
4. The method of claim 1, wherein every stored score is stamped with a frozen norming version such that recomputation under that version reproduces the identical result at any later time.
5. The method of claim 1, wherein all outputs are produced under standing gates comprising educational, non-clinical outputs; input disclosure; and no medical claims.
6. A system comprising one or more processors and memory storing instructions that, when executed, cause the system to perform the method of claim 1.
7. A non-transitory computer-readable medium storing instructions that, when executed by one or more processors, cause performance of the method of claim 1.

Internal Patentability Readiness Assessment

Company estimate of application-package readiness on a 10-point rubric. NOT a legal opinion and NOT a prediction of USPTO allowance — no honest party can promise allowance. Counsel's prior-art search may change any axis.

- Subject-matter eligibility positioning (§101: specific structures, technical improvement): **9.8**
- Enablement and written description (§112: parameters, module paths): **9.8**
- Reduction to practice: **10.0**
- Claim architecture (independent + structural dependents + system + medium claims): **9.6**
- Documented differentiation for prior-art analysis: **9.4**

- **Overall readiness: 9.7 / 10**

- Basis for reduction-to-practice score: Implemented and operating in the JoinAQAL production codebase.

Disclaimer

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DRAFT PROVISIONAL PATENT APPLICATION — JQ-L1-08

Title of Invention: Complementary Cognitive Matching

Series: Series JQ-L1 — JoinAQUAL Founding Patent (Level One; no combinations)

Inventor: Samuel Andrew Russell V

Assignee: Russell Holdings Management, LLC, Wilmington, Delaware

Status: DRAFT — NOT FILED. No application number exists; nothing herein is patent-pending. Per company records; confirm status with patent counsel.

Field of the Invention

Computer-implemented psychometric assessment and personal-development systems; tamper-evident audit of machine-generated assessment.

Summary of the Invention

The invention matches members over their fixed-width 32-dimension vectors under selectable strategies (similarity or complementarity, where one member's demonstrated strength covers another's weakness), and commits every served match set to the audit ledger so match history is tamper-evident.

Abstract

Complementary Cognitive Matching: a computer-implemented system that matches members over their fixed-width 32-dimension vectors under selectable strategies (similarity or complementarity, where one member's demonstrated strength covers another's weakness), and commits every served match set to the audit ledger so match history is tamper-evident.

Technical Improvement (§101 Positioning)

The claims are directed not to an abstract idea but to a specific technical improvement in the functioning of the computing system: matching over measured, contract-fixed cognitive vectors with an auditable serving history, rather than over self-described tags. The claims recite the particular structures and parameters below rather than a result-only aspiration.

Detailed Description

The engine matches members over their fixed-width 32-dimension vectors under selectable strategies (similarity or complementarity, where one member's demonstrated strength covers another's weakness), and commits every served match set to the audit ledger so match history is tamper-evident.

Implementation: matches procedure in server/routers.ts; ledger kind 'match'.

Operating Parameters (Enablement Detail)

Vector-space distance over the 32-dim contract; complementary strategy weights coverage of the counterpart's weak lines; every served set ledger-committed.

Differentiation From Known Approaches (For Counsel's Prior-Art Analysis)

The following contrast reflects the applicant's engineering understanding, not an assertion that no prior art exists: Tag- and questionnaire-based matching: no fixed measurement contract, no verifiable serving history.

Implementation Status

IMPLEMENTATION STATUS (honest enablement). Implemented and operating as a software embodiment in the JoinAQAL codebase at the modules named herein; exercised by the production scoring path and the automated test suite. Hardware embodiments named in related specifications (FPGA co-processing, secure enclaves, HSM ledger anchoring) are NOT built and are not claimed as operating.

Safety and Compliance

SAFETY AND COMPLIANCE GATES. Outputs are educational and non-clinical; all inputs to each result are disclosed to the member; no medical, diagnostic, or treatment claims are made; voice-derived analysis is disclosed on-page before capture and audio is processed in-process on the server, never transmitted to third parties for this analysis.

Claims (DRAFT — for attorney revision)

1. A computer-implemented method of improving the functioning of a cognitive-assessment computing system, the method comprising: (a) matches members over their fixed-width 32-dimension vectors under selectable strategies (similarity or complementarity, where one member's demonstrated strength covers another's weakness), and commits every served match set to the audit ledger so match history is tamper-evident; and (b) presenting the result with a disclosure of the assumptions and inputs that produced it.
2. The method of claim 1, further comprising committing the result to an append-only hash-chained audit ledger whose validity is verifiable end to end by recomputing every link, the chain head being exportable for external anchoring.
3. The method of claim 1, wherein profiles are stored under a fixed-width 32-dimension vector contract with unique indices and scores on [0,1], violating writes being rejected.
4. The method of claim 1, wherein every stored score is stamped with a frozen norming version such that recomputation under that version reproduces the identical result at any later time.
5. The method of claim 1, wherein all outputs are produced under standing gates comprising educational, non-clinical outputs; input disclosure; and no medical claims.
6. A system comprising one or more processors and memory storing instructions that, when executed, cause the system to perform the method of claim 1.
7. A non-transitory computer-readable medium storing instructions that, when executed by one or more processors, cause performance of the method of claim 1.

Internal Patentability Readiness Assessment

Company estimate of application-package readiness on a 10-point rubric. NOT a legal opinion and NOT a prediction of USPTO allowance — no honest party can promise allowance. Counsel's prior-art search may change any axis.

- Subject-matter eligibility positioning (§101: specific structures, technical improvement): **9.8**
- Enablement and written description (§112: parameters, module paths): **9.8**
- Reduction to practice: **10.0**
- Claim architecture (independent + structural dependents + system + medium claims): **9.6**
- Documented differentiation for prior-art analysis: **9.4**
- **Overall readiness: 9.7 / 10**
- Basis for reduction-to-practice score: Implemented and operating in the JoinAQAL production codebase.

Disclaimer

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DRAFT PROVISIONAL PATENT APPLICATION — JQ-L1-09

Title of Invention: Versioned Population Norming with Public Changelog

Series: Series JQ-L1 — JoinAQAL Founding Patent (Level One; no combinations)

Inventor: Samuel Andrew Russell V

Assignee: Russell Holdings Management, LLC, Wilmington, Delaware

Status: DRAFT — NOT FILED. No application number exists; nothing herein is patent-pending. Per company records; confirm status with patent counsel.

Field of the Invention

Computer-implemented psychometric assessment and personal-development systems; tamper-evident audit of machine-generated assessment.

Summary of the Invention

The invention freezes each population-norming curve as an immutable versioned snapshot; every stored score is stamped with its norming version so recomputation under that version reproduces the identical rarity at any later time; a public changelog enumerates every version and an as-of-version endpoint recomputes rarities under any frozen snapshot without mutating stored data.

Abstract

Versioned Population Norming with Public Changelog: a computer-implemented system that freezes each population-norming curve as an immutable versioned snapshot; every stored score is stamped with its norming version so recomputation under that version reproduces the identical rarity at any later time; a public changelog enumerates every version and an as-of-version...

Technical Improvement (§101 Positioning)

The claims are directed not to an abstract idea but to a specific technical improvement in the functioning of the computing system: reproducibility as an architectural guarantee: any historical score can be traced and re-derived under the exact population model that produced it — regulatory-grade transparency. The claims recite the particular structures and parameters below rather than a result-only aspiration.

Detailed Description

The engine freezes each population-norming curve as an immutable versioned snapshot; every stored score is stamped with its norming version so recomputation under that version reproduces the identical rarity at any later time; a public changelog enumerates every version and an as-of-version endpoint recomputes rarities under any frozen snapshot without mutating stored data.

Implementation: server/scoring/norming.ts; GET /api/norms/changelog; GET /api/norms/recompute; norm_version ledger entries.

Operating Parameters (Enablement Detail)

Snapshot = {version, method, description, frozen score-to-rarity curve}; active-version boot entry appended to the ledger only on change; historical scores never re-normed in place.

Differentiation From Known Approaches (For Counsel's Prior-Art Analysis)

The following contrast reflects the applicant's engineering understanding, not an assertion that no prior art exists: Instruments that silently re-norm: historical scores become irreproducible; no public version record.

Implementation Status

IMPLEMENTATION STATUS (honest enablement). Implemented and operating as a software embodiment in the JoinAQAL codebase at the modules named herein; exercised by the production scoring path and the automated test suite. Hardware embodiments named in related specifications (FPGA co-processing, secure enclaves, HSM ledger anchoring) are NOT built and are not claimed as operating.

Safety and Compliance

SAFETY AND COMPLIANCE GATES. Outputs are educational and non-clinical; all inputs to each result are disclosed to the member; no medical, diagnostic, or treatment claims are made; voice-derived analysis is disclosed on-page before capture and audio is processed in-process on the server, never transmitted to third parties for this analysis.

Claims (DRAFT — for attorney revision)

1. A computer-implemented method of improving the functioning of a cognitive-assessment computing system, the method comprising: (a) freezes each population-norming curve as an immutable versioned snapshot; every stored score is stamped with its norming version so recomputation under that version reproduces the identical rarity at any later time; a public changelog enumerates every version and an as-of-version endpoint recomputes rarities under any frozen snapshot without mutating stored data; and (b) presenting the result with a disclosure of the assumptions and inputs that produced it.
2. The method of claim 1, further comprising committing the result to an append-only hash-chained audit ledger whose validity is verifiable end to end by recomputing every link, the chain head being exportable for external anchoring.
3. The method of claim 1, wherein profiles are stored under a fixed-width 32-dimension vector contract with unique indices and scores on [0,1], violating writes being rejected.
4. The method of claim 1, wherein every stored score is stamped with a frozen norming version such that recomputation under that version reproduces the identical result at any later time.
5. The method of claim 1, wherein all outputs are produced under standing gates comprising educational, non-clinical outputs; input disclosure; and no medical claims.
6. A system comprising one or more processors and memory storing instructions that, when executed, cause the system to perform the method of claim 1.
7. A non-transitory computer-readable medium storing instructions that, when executed by one or more processors, cause performance of the method of claim 1.

Internal Patentability Readiness Assessment

Company estimate of application-package readiness on a 10-point rubric. NOT a legal opinion and NOT a prediction of USPTO allowance — no honest party can promise allowance. Counsel's prior-art search may change any axis.

- Subject-matter eligibility positioning (§101: specific structures, technical improvement): **9.8**
- Enablement and written description (§112: parameters, module paths): **9.8**
- Reduction to practice: **10.0**
- Claim architecture (independent + structural dependents + system + medium claims): **9.6**
- Documented differentiation for prior-art analysis: **9.4**

- **Overall readiness: 9.7 / 10**

- Basis for reduction-to-practice score: Implemented and operating in the JoinAQAL production codebase.

Disclaimer

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DRAFT PROVISIONAL PATENT APPLICATION — JQ-L1-10

Title of Invention: Sovereign Identity Passport with Per-Line Research Provenance

Series: Series JQ-L1 — JoinAQUAL Founding Patent (Level One; no combinations)

Inventor: Samuel Andrew Russell V

Assignee: Russell Holdings Management, LLC, Wilmington, Delaware

Status: DRAFT — NOT FILED. No application number exists; nothing herein is patent-pending. Per company records; confirm status with patent counsel.

Field of the Invention

Computer-implemented psychometric assessment and personal-development systems; tamper-evident audit of machine-generated assessment.

Summary of the Invention

The invention consolidates a member's scores, verified floors, controlling weakness, norming version, and per-line research provenance into one exportable identity document (plain JSON and JSON-LD with schema.org annotations), served under scoped read-only slices, with EVERY access — including the member's own — written to an append-only access log the member can audit; each of the 32 lines is mapped at the database level to the named research tradition and scholars grounding it, DOI fields left null rather than invented.

Abstract

Sovereign Identity Passport with Per-Line Research Provenance: a computer-implemented system that consolidates a member's scores, verified floors, controlling weakness, norming version, and per-line research provenance into one exportable identity document (plain JSON and JSON-LD with schema.org annotations), served under scoped read-only slices, with EVERY access — including...

Technical Improvement (§101 Positioning)

The claims are directed not to an abstract idea but to a specific technical improvement in the functioning of the computing system: data sovereignty as mechanism: the member holds one portable, machine-readable, provenance-grounded identity document and an audit trail of everyone who read it. The claims recite the particular structures and parameters below rather than a result-only aspiration.

Detailed Description

The engine consolidates a member's scores, verified floors, controlling weakness, norming version, and per-line research provenance into one exportable identity document (plain JSON and JSON-LD with schema.org annotations), served under scoped read-only slices, with EVERY access — including the member's own — written to an append-only access log the member can audit; each of the 32 lines is mapped at the database level to the named research tradition and scholars grounding it, DOI fields left null rather than invented.

Implementation: sovereign router in server/routers.ts; server/patents/provenanceSeed.ts; identity_access_log (migration 0031).

Operating Parameters (Enablement Detail)

Scopes: full | rarity | weakness | floors; JSON-LD Person/PropertyValue rendering; idempotent 32-row provenance seed at boot; access log rows {userId, scope, endpoint, accessor, timestamp}.

Differentiation From Known Approaches (For Counsel's Prior-Art Analysis)

The following contrast reflects the applicant's engineering understanding, not an assertion that no prior art exists: Platform-locked profiles with dataset-level provenance and no member-visible access accounting.

Implementation Status

IMPLEMENTATION STATUS (honest enablement). Implemented and operating as a software embodiment in the JoinAQAL codebase at the modules named herein; exercised by the production scoring path and the automated test suite. Hardware embodiments named in related specifications (FPGA co-processing, secure enclaves, HSM ledger anchoring) are NOT built and are not claimed as operating.

Safety and Compliance

SAFETY AND COMPLIANCE GATES. Outputs are educational and non-clinical; all inputs to each result are disclosed to the member; no medical, diagnostic, or treatment claims are made; voice-derived analysis is disclosed on-page before capture and audio is processed in-process on the server, never transmitted to third parties for this analysis.

Claims (DRAFT — for attorney revision)

1. A computer-implemented method of improving the functioning of a cognitive-assessment computing system, the method comprising: (a) consolidates a member's scores, verified floors, controlling weakness, norming version, and per-line research provenance into one exportable identity document (plain JSON and JSON-LD with schema.org annotations), served under scoped read-only slices, with EVERY access — including the member's own — written to an append-only access log the member can audit; each of the 32 lines is mapped at the database level to the named research tradition and scholars grounding it, DOI fields left null rather than invented; and (b) presenting the result with a disclosure of the assumptions and inputs that produced it.
2. The method of claim 1, further comprising committing the result to an append-only hash-chained audit ledger whose validity is verifiable end to end by recomputing every link, the chain head being exportable for external anchoring.
3. The method of claim 1, wherein profiles are stored under a fixed-width 32-dimension vector contract with unique indices and scores on [0,1], violating writes being rejected.
4. The method of claim 1, wherein every stored score is stamped with a frozen norming version such that recomputation under that version reproduces the identical result at any later time.
5. The method of claim 1, wherein all outputs are produced under standing gates comprising educational, non-clinical outputs; input disclosure; and no medical claims.
6. A system comprising one or more processors and memory storing instructions that, when executed, cause the system to perform the method of claim 1.
7. A non-transitory computer-readable medium storing instructions that, when executed by one or more processors, cause performance of the method of claim 1.

Internal Patentability Readiness Assessment

Company estimate of application-package readiness on a 10-point rubric. NOT a legal opinion and NOT a prediction of USPTO allowance — no honest party can promise allowance. Counsel's prior-art search may change any axis.

- Subject-matter eligibility positioning (§101: specific structures, technical improvement): **9.8**
- Enablement and written description (§112: parameters, module paths): **9.8**
- Reduction to practice: **10.0**

- Claim architecture (independent + structural dependents + system + medium claims): **9.6**
- Documented differentiation for prior-art analysis: **9.4**
- **Overall readiness: 9.7 / 10**
- Basis for reduction-to-practice score: Implemented and operating in the JoinAQAL production codebase.

Disclaimer

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